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Long-Term Continuous Monitoring of New-Onset Atrial Fibrillation After Coronary Artery Bypass Grafting

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 Editorial

 Supplemental content

IMPORTANCE The incidence and burden of new-onset atrial fibrillation (AF) after coronary artery bypass grafting (CABG) are not known. Nevertheless, North American guidelines state that it is reasonable to administer 60 days of oral anticoagulation to patients with new-onset AF after CABG, a moderate-strength recommendation (class 2a) based on evidence derived from nonrandomized clinical studies.

OBJECTIVE To test the hypothesis that the incidence of new-onset AF within the first year after CABG is higher than suggested in the current literature and to assess AF burden.

DESIGN, SETTING, AND PARTICIPANTS A prospective, multicenter cohort study in 2 academic cardiac surgery centers in Germany that involved 198 patients with 3-vessel coronary artery disease or left main disease and no history of arrhythmias who received an insertable cardiac monitor during CABG for long-term continuous electrocardiographic monitoring. Patients were enrolled from November 2019 through November 2023 and were followed up for 1 year.

EXPOSURES Atrial fibrillation detected within a year of continuous monitoring using insertable cardiac monitors implanted during CABG.

MAIN OUTCOMES AND MEASURES Cumulative incidence of new-onset AF within a year of surgery. Secondary outcomes were AF burden and clinical outcomes.

RESULTS A total of 1217 patients were assessed, and 1008 were excluded. Of the 198 patients enrolled (173 male [87.4%]; 25 female [12.6%]; mean age, 66 [SD, 9] years), 95 patients developed new-onset AF within the first year after CABG (cumulative incidence, 48%; 95% CI, 41%-55%). The median AF burden during the first year was 0.07% (IQR, 0.02%-0.23%) or 370 minutes. The median AF burden on days 1 through 7 was 3.65% (IQR, 0.95%-9.09%); on days 8 through 30, 0.04% (IQR, 0%-1.21%); and on days 31 through 365, 0% (IQR, 0%-0.0003%), corresponding to 368, 13, and 0 minutes, respectively. After discharge, 3 patients had an AF episode longer than 24 hours.

CONCLUSIONS AND RELEVANCE Although the incidence of new-onset AF after CABG in this study was higher than previously reported, the AF burden in these patients was very low, especially after 30 days. The very low AF burden questions the current guideline recommendations that long-term oral anticoagulation should be considered in patients with new-onset AF after CABG.

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New-onset postoperative atrial fibrillation (AF) after coronary artery bypass grafting (CABG) is the most frequent early complication after surgery with a reported incidence of approximately 30%.^{1,2} New-onset AF after CABG is associated with prolonged postoperative hospitalization, increased costs, and discomfort for the patient and is, according to observational data, associated with an increased long-term risk of thromboembolic stroke,³ heart failure, and AF recurrence.² Current North American guidelines state that “it is reasonable to administer anticoagulation for 60 days after surgery and to reevaluate the need for longer term anticoagulation,”⁴ and according to European guidelines long-term anticoagulation should be considered for patients with new-onset AF after cardiac surgery,⁵ both moderate-strength recommendations (class 2a) based on evidence derived from nonrandomized studies).

Large-scale contemporary studies investigating new-onset AF after cardiac surgery have been limited methodologically by their temporally restricted application of in-hospital telemetry and intermittent short-term monitoring after discharge.^{6,7} These limitations prevent an accurate estimation of the incidence of new-onset AF after CABG and furthermore leave the issues of AF burden and recurrence unknown. There has been a call for investigations of AF after CABG using long-term continuous monitoring to close this gap in knowledge.^{3,8} Few studies have applied long-term continuous monitoring to investigate AF after cardiac surgery. These studies are limited by their small cohort sizes,⁹ their investigation of mixed cardiac surgery cohorts,^{10,11} monitors implanted after AF has been detected,¹¹⁻¹³ and inclusion of patients with a history of AF.¹⁴ Thus far, only one small study has investigated the development of AF after CABG by implanting insertable cardiac monitors (ICMs) at the time of surgery.¹⁵

A prospective observational study with continuous monitoring was performed to test the hypothesis that the incidence of new-onset AF within the first year after CABG is significantly higher than the incidence suggested in the literature and to assess AF burden.

Methods

Study Design

In this investigator-initiated prospective observational study, Characterization of Post Coronary Artery Bypass Grafting Atrial Fibrillation Patterns/CABG-AF, with no industry funding, ICMs (Reveal LINQ LNQ11, Medtronic) were implanted in patients at the time of CABG (for the trial protocol see [Supplement 1](#)). ICMs are 4.5 × 0.7 × 0.4-cm capsules implanted in the subcutaneous tissue of the left chest through an incision that is less than 1 cm. The devices detect and store details of rhythm irregularities; data are regularly transferred to a database using a bedside unit operated by the patient (MyCareLink home monitor, Medtronic). Study rationale and design have previously been described.¹⁶ The study was performed at 2 cardiac surgery centers in Germany, LMU University Hospital, Munich, and Jena University Hospital, Jena. Institutional review boards of both sites approved the study. This study followed

Key Points

Question What are the incidence and burden of atrial fibrillation (AF) after coronary artery bypass grafting (CABG) as measured by continuous cardiac monitoring?

Findings In this prospective cohort study of 198 patients who received an insertable cardiac monitor implant during CABG, 48% developed new-onset AF within 1 year of surgery. The median AF burden in the first year after surgery was 0.07%.

Meaning The incidence of new-onset AF after CABG is markedly higher than previously reported; however, the AF burden after surgery is very low, questioning current guidelines recommending oral anticoagulation for new-onset AF after CABG.

the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

Study Population

Eligible patients were adults (>18 years) undergoing first-time isolated CABG for 3-vessel coronary artery or left main disease receiving 2 or more bypass grafts, with no history of AF or other arrhythmias, and a preoperative left ventricular ejection fraction (LVEF) of 35% or more. Patients requiring concomitant surgical procedures or mechanical circulatory support when weaning from cardiopulmonary bypass were excluded. Eligibility criteria are summarized in [eTable 1](#) in [Supplement 2](#). All patients provided written informed consent.

Study Procedure

After skin closure, an ICM was implanted. Continuous monitoring was initiated in the operating room. Upon detection of arrhythmias, patients and treating physicians were informed. Subsequent clinical decisions (eg, oral anticoagulation) were made by the treating physicians. Clinical follow-up data were collected by mailed follow-up forms and telephone interviews.

Outcome Measures

The primary outcome was the occurrence of new-onset AF detected by continuous monitoring within the first year after CABG. Secondary outcomes were total time in AF, AF burden (a measure of the amount of time in AF in a specified monitoring period, expressed as a percentage),¹⁷ prevalence of episodes not detected by standard monitoring, prevalence of asymptomatic episodes, and major adverse cardiac and cardiovascular events and their components ([eTable 2](#) in [Supplement 2](#)).

AF was defined as any arrhythmia episode detected using the cardiac monitor's algorithms (AF detection, atrial tachycardia detection, tachycardia detection) and adjudicated as AF by the adjudication team (consisting of a cardiac electrophysiologist and a cardiac surgeon; [eMethods 1](#) and [eResults 1](#) in [Supplement 2](#)). AF was identified by the device at a duration of 2 or more minutes.

After each episode or episode cluster, the adjudication team evaluated whether AF had been detected by standard monitoring by interviewing the patient or the patient's physician

and reviewing clinical documentation. Standard monitoring, in accordance with local hospital protocol, consisted of 2 to 5 days of postoperative telemetry monitoring and a 12-lead electrocardiogram (ECG) prior to discharge. Some patients underwent further investigation, such as extension of telemetry, Holter monitoring, and further 12-lead ECGs, as determined by treating physicians.

Statistical Analysis

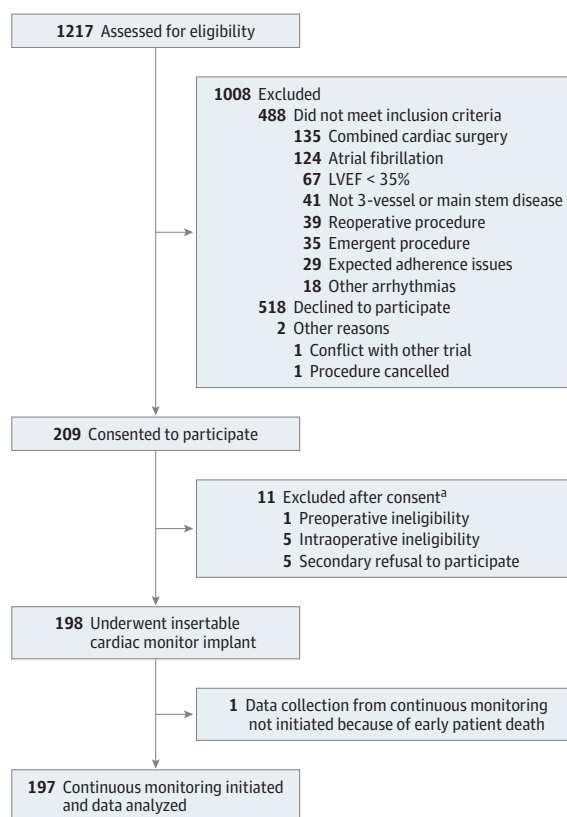
The primary hypothesis was that the incidence of AF detected with continuous monitoring within the first year after CABG is significantly higher than the incidence suggested by current published data ($\approx 30\%$).^{1,2} Planning to calculate the incidence within 1 year of CABG with a 2-sided 95% CI with a prespecified width of no more than plus or minus 7% and assuming that so few deaths occur within the study time frame that they may be neglected, the required sample size, calculated with nQuery 8.1.2.0 (Statistical Solutions), was 196. To account for withdrawal and loss to follow-up, the sample size was increased to 198. All patients with an implanted monitor constitute the intention-to-monitor (ITM) population. All analyses were performed on the ITM population.

The primary outcome was investigated using a cumulative incidence function considering the competing risk of death. Post hoc sensitivity analyses for the primary outcome were performed based on altered temporal thresholds for defining AF (cumulative time in AF: 4, 6, and 12 minutes instead of 2 minutes as in the primary analysis). Worst-case analyses, for which the only missing case was first assumed to have developed AF and second assumed not to have, were performed to test the robustness of the results. The cumulative incidence detected with standard monitoring served as a substitute for the incidence reported in the literature in which most studies describe similar methods of monitoring and monitoring times. The cumulative incidence detected by continuous monitoring was compared with the incidence detected with standard monitoring using Gray's test.

Cumulative time in AF was calculated by adding the duration of all adjudicated episodes. In line with the 2025 European Society of Cardiology (ESC) consensus on AF burden,¹⁷ burden was calculated by dividing cumulative time in AF by total time monitored.^{5,18} For AF burden graphs, the mean daily AF burden was calculated by dividing the cohort's cumulative time in AF on a given day by the cumulative number of patients who had developed AF up to this time point.

Categorical variables are reported as counts (%); continuous variables as mean (SD) or median (IQR) in accordance with data normality tests. Differences between patients with and without AF were assessed using the χ^2 test or Fisher exact test for categorical variables and *t* test or the Wilcoxon rank-sum test for continuous variables (as appropriate). *P* values are 2-sided, with an α level of .05 indicating statistical significance. Statistics were performed using R version 4.2.2 (R Foundation for Statistical Computing) with additional packages (eTable 3 in Supplement 2). Graphs were prepared using R and GraphPad Prism version 10.1.2 (GraphPad Software Inc).

Figure 1. Flowchart of Assessment and Enrollment of Patients in the CABG-AF Study



^aOf those excluded after providing consent, 1 patient had a history of arrhythmias; 5 patients were ineligible intraoperatively (2 due to an expansion of the procedure to encompass more than coronary artery bypass grafting, 2 due to the intraoperative implant of an intra-aortic balloon pump, and 1 due to an intraoperative exclusion criterion that was not documented in detail); and 5 patients withdrew consent.

CABG-AF indicates Characterization of Post Coronary Artery Bypass Grafting Atrial Fibrillation Patterns study; and LVEF, left ventricular ejection fraction.

Results

Participant Characteristics

From November 2019 through November 2023, 1217 patients were assessed for eligibility, and 198 patients (173 male [87.4%]; 25 female [12.6%]; mean age, 66 years [SD, 9]) were enrolled and had an ICM implanted (Figure 1; eTables 4 and 5 in Supplement 2). Race and ethnicity were not collected.

Remote data collection was initiated for 197 patients (99.5%); 1 patient (0.5%) died before initiation. One year of continuous monitoring was completed for 192 patients (97.0%). Reasons for not completing 1 year of monitoring included 4 (2.0%) deaths, 1 (0.5%) withdrawal of consent, and 1 (0.5%) loss to follow-up. The complete monitoring time for the cohort was 193 patient-years, and the mean monitoring time was 0.975 years.

Baseline and surgical characteristics are presented in Table 1. In the ITM population, the median CHA₂DS₂-VASc (congestive heart failure, hypertension, age ≥ 75 years, diabetes

Table 1. Patient Baseline and Surgical Characteristics, Intention-to-Monitor Population^a

	Group, No. (%) of patients	
	New-onset AF (n = 95) ^b	No new-onset AF (n = 103)
Sex		
Female	14 (14.7)	11 (10.7)
Male	81 (85.3)	92 (89.3)
Age, mean (SD), y	68 (8)	63 (10)
Hypertension	79 (83.2)	91 (88.3)
Dyslipidemia	74 (77.9)	76 (73.8)
Diabetes mellitus	35 (36.8)	38 (36.9)
Peripheral vascular disease	9 (9.5)	8 (7.8)
COPD	4 (4.2)	5 (4.9)
Previous ischemic stroke or TIA	5 (5.3)	6 (5.8)
CHA ₂ DS ₂ -VASc score, median (IQR) ^c	3 (2-4)	3 (2-4)
EuroSCORE II, median (IQR) ^d	1.01 (0.75-1.39)	0.79 (0.66-1.20)
NYHA functional classification		
I (Least severe)	53 (55.8)	59 (57.3)
II (Mildly severe)	28 (29.5)	35 (34.0)
III (Moderately severe)	13 (13.7)	9 (8.7)
IV (Most severe)	1 (1.1)	0
Previous acute MI	26 (27.4)	32 (31.1)
Previous PCI	21 (22.1)	24 (23.3)
Left ventricular ejection fraction, %		
≥55	74 (77.9)	76 (73.8)
45-54	17 (17.9)	22 (21.4)
35-44	4 (4.2)	5 (4.9)
Left ventricular hypertrophy	43 (45.3)	28 (27.2)
Chronic kidney failure (eGFR <60 mL/min/1.73 m ²)	15 (15.8)	9 (8.7)
Three-vessel disease	88 (92.6)	97 (94.2)
Left main disease	42 (44.2)	49 (47.6)
Proximal left anterior descending stenosis	69 (72.6)	71 (68.9)
On-pump CABG, median (IQR), min	77 (81.1)	74 (71.8)
Cardiopulmonary bypass time	108 (93-129)	104 (88-126)
Aortic cross-clamp time	70 (60-85)	64 (57-83)
Off-pump CABG	18 (18.9)	29 (28.2)
Multiple arterial grafting	55 (57.9)	66 (64.1)
Single arterial grafting	40 (42.1)	37 (35.9)
Use of left internal mammary artery	91 (95.8)	99 (96.1)
No. of peripheral anastomoses		
2	16 (16.8)	25 (24.3)
3	56 (58.9)	62 (60.2)
4	21 (22.1)	14 (13.6)
5	2 (2.1)	2 (1.9)
Postoperative stay, median (IQR), d		
Intensive care unit	3 (2-4)	2 (2-4)
Hospital	10 (8-13)	9 (7-11)

Abbreviations: CABG, coronary artery bypass grafting; COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate; EuroSCORE, European System for Cardiac Operative Risk Evaluation; MI, myocardial infarction; NYHA, New York Heart Association; PCI, percutaneous coronary intervention; TIA, transient ischemic attack.

^a Baseline characteristics were determined at the time of surgery, surgical characteristics directly after surgery. No baseline or surgical data were missing.

^b Occurrence of new-onset atrial fibrillation (AF) within the first year after CABG.

^c CHA₂DS₂-VASc score (congestive heart failure, hypertension, age ≥75 years, diabetes mellitus, stroke or transient ischemic attack or thromboembolism, vascular disease, age 65-74 years, and female sex); higher scores indicate a higher thromboembolic risk, a median score of 3 indicates a moderate to high risk.

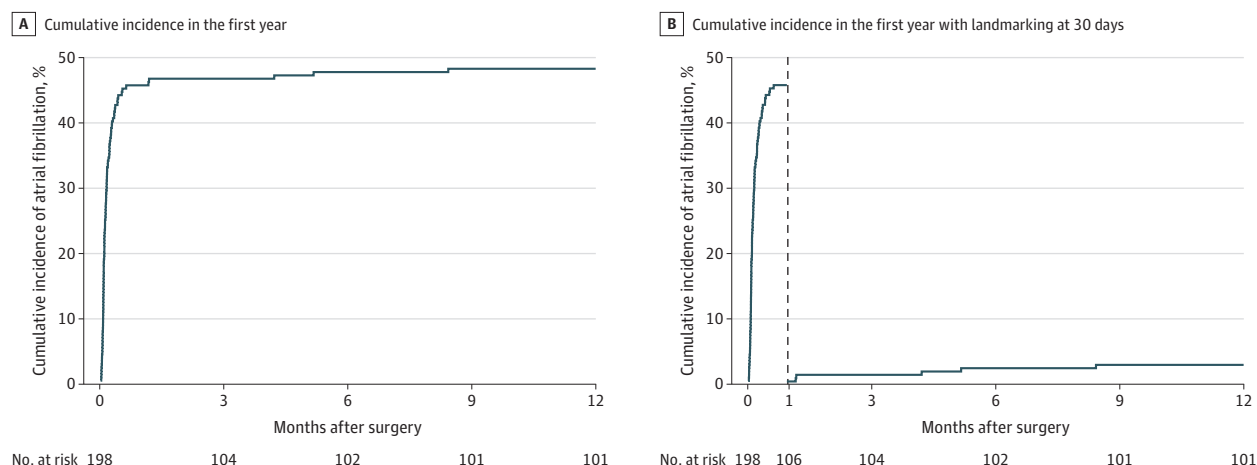
^d EuroSCORE, a risk model used to estimate in-hospital mortality in patients who have undergone cardiac surgery, including patient-, cardiac-, and operation-related factors (a median score of ≈ 1% indicates a low predicted risk of in-hospital mortality).

mellitus, stroke or transient ischemic attack or thromboembolism, vascular disease, age 65-74 years, and female sex) score was 3 (IQR, 2-4; eTable 6 in [Supplement 2](#)), 58% of patients received β-blockers, and no patient received amiodarone before surgery. The median postoperative hospital stay was 10 days (IQR, 8-11).

Primary Outcome

Ninety-five patients in the ITM population developed new-onset AF, the primary end point. The cumulative incidence of new-onset AF within 1 year after surgery was 48% (95% CI, 41%-55%; [Figure 2](#); eTable 7 in [Supplement 2](#)). In sensitivity analyses, applying AF thresholds for cumulative time in AF, the cu-

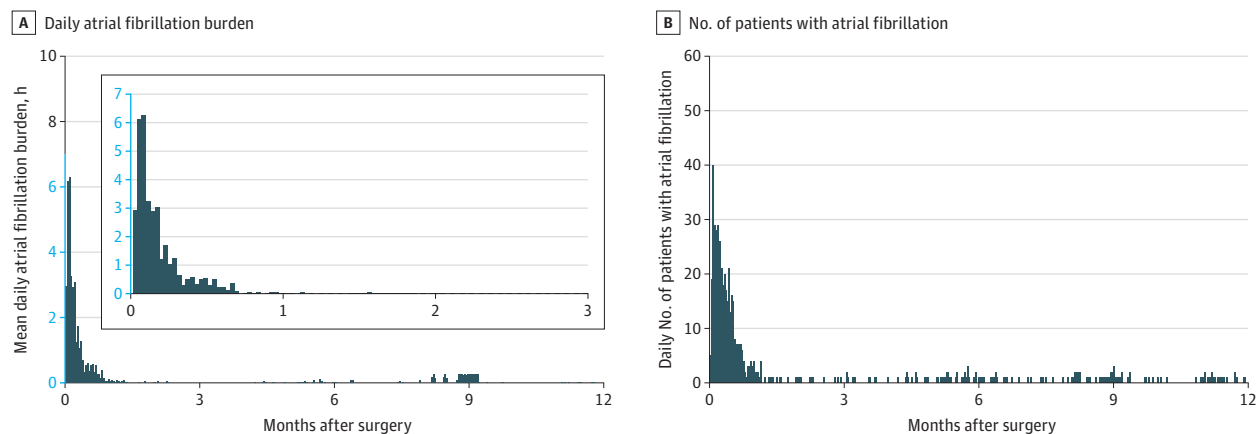
Figure 2. Cumulative Incidence of New-Onset Atrial Fibrillation After Coronary Artery Bypass Grafting



The graphs represent cumulative incidence functions considering death as a competing risk, investigating the intention-to-monitor population. A and B present the same data; in panel B, the graph was split with a landmark at 30

days (dotted line) to illustrate the low incidence in the time frame later than 30 days after surgery.

Figure 3. Atrial Fibrillation Burden in Patients With New-Onset Atrial Fibrillation After Coronary Artery Bypass Grafting



A, The mean daily atrial fibrillation (AF) burden was calculated by dividing the cohort's cumulative time in AF on a given day by the cumulative number of patients who had developed AF up to this time point. The inset shows the same data on enlarged y-axis and x-axis. Note that the maximum possible AF burden

on a given day is 24 hours. Panel B depicts the number of patients with AF on a given day (irrespective of AF burden on the specific day). The median length of hospital stay for the patients was 10 (IQR, 8-13) days.

cumulative incidences were 46% (95% CI, 38%-52%) for 4 minutes, 45% (95% CI, 37%-51%) for 6 minutes; and 44% (95% CI, 37%-50%) for 12 minutes; eTables 8-10 in [Supplement 2](#). Worst-case analyses had no effect on the results (eResults 2 in [Supplement 2](#)).

Secondary Outcomes

A total of 2053 individual AF episodes constituting 2522 hours occurred within the monitored time. The median AF burden within the first year in patients with new-onset AF was 0.07% (IQR, 0.02%-0.23%) corresponding to 370 minutes of cumulative time in AF (**Figure 3**). The median AF burden on days 1 through 7 was 3.65% (IQR, 0.95%-9.09%); 8 through 30, 0.04% (IQR, 0%-1.21%); and 31 through 365,

0% (IQR, 0%-0.0003%). Patient-specific burden data are demonstrated in heat maps (eFigures 1 and 2 in [Supplement 2](#)). The median time to incident AF episode was 3.3 days (IQR, 2.1-6.6). The median episode length was 6 minutes (IQR, 2-22). Altogether 63% (95% CI, 61%-66%) of episodes were asymptomatic and 67% (95% CI, 65%-69%) were not detected by standard monitoring. The cumulative incidence of AF detected with standard monitoring was 34% (95% CI, 27%-41%; Gray's test $P = .01$ vs continuous monitoring) (eTable 11 in [Supplement 2](#)).

Among 73 of 95 patients who developed AF (77%; 95% CI, 67%-85%), the incident AF episode occurred within the first 7 days; the highest incidence was between 24 and 72 hours after surgery. Forty-five percent of the time that patients were

Table 2. Clinical Outcomes During the First Year After Coronary Artery Bypass Grafting, Intention-to-Monitor Population

	Group, No. (%) of patients		Estimated risk difference (95% CI) ^c	P value ^d
	New-onset AF (n = 95) ^a	No new-onset AF (n = 103) ^b		
MACE	7 (7.4)	7 (6.8)	0.006 (−0.07 to 0.084)	>.99
Death	3 (3.2)	1 (1.0)	0.023 (−0.023 to 0.075)	.35
Cardiac death	0	1 (1.0)	−0.011 (−0.046 to 0.023)	>.99
Stroke	0	2 (1.9)	−0.021 (−0.062 to 0.016)	.50
Myocardial infarction	1 (1.1)	1 (1.0)	0.001 (−0.039 to 0.043)	>.99
Repeat revascularization	1 (1.1)	4 (3.9)	−0.029 (−0.083 to 0.02)	.37
Heart failure hospitalization	4 (4.2)	0	0.044 (0.004 to 0.098)	.05
Readmission	17 (17.9)	22 (21.4)	−0.035 (−0.146 to 0.078)	.59
Cardiac readmission	6 (6.3)	5 (4.9)	0.015 (−0.054 to 0.087)	.76

Abbreviation: MACE, major adverse cardiac or cardiovascular event, defined as the composite of death, myocardial infarction, coronary revascularization, stroke, and heart failure hospitalization.

^a Occurrence of new-onset atrial fibrillation (AF) within the first year after coronary artery bypass grafting. One patient was censored on day 27 because of withdrawal of consent.

^b One patient was censored on day 19 because of loss to follow-up.

^c Risk differences and corresponding 95% CIs were estimated using the asymptotic score test.

^d P values were calculated using the Fisher exact test.

in AF had occurred within the first 7 days and 77% occurred within the first 30 days after surgery. Among 90 of 95 patients (95%) who developed AF (95% CI, 88%-98%), the incident AF episode occurred within the first 30 days. The 5 patients who had incident episodes after day 30 experienced 78 episodes with a median length of 4 minutes. Median cumulative time in AF for these patients was 50 minutes (IQR, 24-716).

Of the 90 patients with incident episodes prior to 30 days after surgery, 19 patients (21%; 95% CI, 13%-31%) had recurrence later than 30 days. The 554 recurrent episodes occurring later than 30 days after surgery had a median episode length of 4 minutes; 237 were asymptomatic (43%; 95% CI, 39%-47%) and 15 of the 554 recurrent episodes were detected with standard monitoring (3%; 95% CI, 2%-4%).

Ten patients had 15 AF episodes that lasted 24 or more hours (0.7% of all episodes; 95% CI, 0.4%-1.2%). Eight of the 15 episodes lasting 24 or more hours occurred in 8 patients prior to discharge and were detected with both continuous and standard monitoring. Seven episodes lasting 24 or more hours occurred in 3 patients after discharge and were not detected with standard monitoring.

One year after surgery, 7 patients (7.4%) with and 7 (6.8%) without new-onset AF experienced at least 1 major adverse cardiac and cardiovascular event ($P > .99$; Table 2). Three patients (3.2%) with and 1 (1.0%) without AF died during follow-up ($P = .35$). No patient with new-onset AF developed a stroke during follow-up, whereas 2 patients (1.9%) without did develop a stroke ($P = .50$). Both strokes were nondisabling and occurred during the initial hospitalization. At discharge and at 1 year, the proportion of patients treated with oral anticoagulants was higher in patients with new-onset AF. For medication prescription data see eTables 12 and 13 in Supplement 2.

Discussion

This prospective, observational, multicenter study with continuous rhythm monitoring found that although the incidence of AF early after CABG is higher than previously reported, the median AF burden is very low within 30 days and minuscule later than 30 days after surgery.

The incidences of new-onset AF reported after cardiac surgery are dependent on study cohort, method of detection, and duration of monitoring. Both prospective randomized studies using telemetry and contemporary large observational studies have reported incidences of 25% to 30% after CABG,^{2,6,7,19,20} which correlates well with the 27% incidence with standard monitoring 5 days after surgery observed in the present study. Analyzing continuous monitoring data, a significantly higher incidence (48%) was observed, which confirms the hypothesis that standard in-hospital monitoring (as predominantly presented in the literature) underestimates the incidence of new-onset AF after CABG. Continuous monitoring allows analysis of AF patterns.¹⁷ A low burden pattern with a high aggregation of episodes early after surgery was observed. Later than 30 days after surgery, a 21% rate of recurrence was detected. A higher rate has been noted by a study investigating AF recurrence after new-onset AF during medical and noncardiac surgical hospitalizations (33%).²¹ The time frame for detecting new-onset AF, the long duration of AF during hospitalization, and clinical differences provide an explanation for this higher rate.

There is conflicting evidence regarding the link between postoperative AF after cardiac surgery and risk of stroke, with some studies demonstrating no association.²²⁻²⁴ Nevertheless, the 2023 American College of Cardiology (ACC)/American Heart Association (AHA) AF guidelines' moderate-strength recommendation (class 2a) based on nonrandomized studies states that "it is reasonable to administer anticoagu-

lation for 60 days and to reevaluate the need for longer term anticoagulation.”⁴ Likewise, the 2024 joint ESC and European Association for Cardio-Thoracic Surgery AF guidelines’ moderate-strength recommendation (class 2a) based on non-randomized studies states that “long-term anticoagulation should be considered in patients at elevated thromboembolic risk.”⁵ Although in the present study some AF episodes were detected later than 30 days after surgery, the burden after 30 days was very low (median burden, 0%; IQR, 0%-0.0003%). This low burden calls into question whether long-term oral anticoagulation is necessary in patients with new-onset AF after CABG. The very low burden provides a likely explanation for why observational studies have failed to demonstrate reduced stroke rates with oral anticoagulation in this patient group.^{2,25-28} Notably, episodes occurring later than 30 days after CABG were almost all undetected with standard monitoring and can consequently be equated with subclinical AF as defined in the ACC/AHA guidelines.⁴ Recent data demonstrated an overall low stroke rate of around 1% per patient-year among patients with subclinical AF treated with acetylsalicylic acid.²⁹⁻³¹ The Asymptomatic Atrial Fibrillation and Stroke Evaluation in Pacemaker Patients and the Atrial Fibrillation Reduction Atrial Pacing Trial (ASSERT),³² which investigated device-detected AF, provided evidence that only subclinical AF with an episode duration of 24 or more hours was associated with an increased risk of stroke. The ACC/AHA guidelines state that it is reasonable to initiate oral anticoagulation only if subclinical AF with a duration of 24 or more hours has been detected.⁴ Current ESC guidelines, including recent evidence from the Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial High Rate Episodes (NOAH-AFNET 6)²⁹ and Apixaban for the Reduction of Thromboembolism in Patients with Device-Detected Subclinical Atrial Fibrillation (ARTESIA)³¹ trials, have a weak recommendation that oral anticoagulation may be considered for patients with subclinical AF.⁵ In the present study, only 15 of 2053 AF episodes (0.7%) in 10 patients lasted 24 or more hours. Most of these long AF episodes occurred during the index hospitalization when patients are generally monitored with telemetry and are treated with prophylactic low-molecular-weight heparin. In the present study, only 7 episodes among 3 patients lasting 24 or more hours would have gone unidentified with standard monitoring owing to occurrence after discharge.

To date, no randomized studies supporting the use of oral anticoagulants in patients with new-onset AF after CABG have been published. Observational studies and meta-analyses have observed an increased rate of bleeding with no difference in stroke rate in patients with new-onset AF after CABG who were treated with oral anticoagulants.^{2,25-27} The randomized Anticoagulation for New-Onset Post-Operative Atrial Fibrillation After CABG (PACES) trial (NCT04045665) is currently comparing 3 months of single antiplatelet therapy with 3 months of combined oral anticoagulant and antiplatelet therapy in patients with new-onset AF after CABG.

Based on the low burden of AF, the predominance of subclinical AF after discharge, the very low incidence of AF episodes with a duration of 24 or more hours in the present study, the widespread use of low-molecular-weight heparin for thromboembolic prophylaxis before discharge, the current recommendations for anticoagulation in subclinical AF, and observational evidence suggesting a lack of net clinical benefit, the current recommendations for oral anticoagulation in patients with new-onset AF after CABG can be questioned. The data in this current study suggest that if oral anticoagulation is initiated after surgery in patients with new-onset AF, a reevaluation of the need for its continuation should be performed 30 days after surgery.

Limitations

Although the strength of this study is its continuous, meticulous long-term monitoring, there are several limitations. First, the mechanism of AF investigation in 2-minute time frames infers that episodes less than 2 minutes long may go undetected and that short time frames of sinus rhythm may lead to episode splitting, with a subsequent effect on episode number and length. Second, because of limited data storage capacity, the occurrence of new episodes prior to the transfer of stored episodes led to episode overwriting in 5 patients. A recalculation of the AF burden in these patients, based on the ICM’s daily time in AF trend data rather than based on adjudicated episodes, only had a small effect on the median AF burden (0.08% vs 0.07%). Third, the nonblinded design of the study may have had an effect on whether individual episodes were detected by standard monitoring. Fourth, the relatively small sample size and the generalizability represent further limitations. The conclusions of this study are applicable only to the studied population, patients who had undergone CABG with 3-vessel or left main disease, a left ventricular ejection fraction of 35% or more, and no prior AF. The study included elective patients with predominantly low surgical risk. Fifth, this study did not assess the value of anticoagulation in patients with AF after CABG. Furthermore, no determinations can be made regarding the utility of continuous monitoring for guidance of anticoagulation in this population.

Conclusions

Although the incidence of new-onset AF after CABG is higher than previously reported, the burden both within the first 30 days and later than 30 days after surgery is very low. Episodes with a duration of 24 or more hours are very rare. After discharge, AF is largely subclinical and of short duration. Taken together with previous investigations, these data question the current recommendations for oral anticoagulation in patients with new-onset AF after CABG. If oral anticoagulation is initiated, a reevaluation 30 days after surgery is recommended.

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Author Contributions: Drs Herrmann and Hagl had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Herrmann, Brömsen, Massberg, Doenst, Juchem, Hagl.

Acquisition, analysis, or interpretation of data: Herrmann, Jeppsson, Kirov, Charitos, Dacian, Massberg, Sadoni, Hagl.

Drafting of the manuscript: Herrmann.

Critical review of the manuscript for important intellectual content: All authors.

Statistical analysis: Herrmann.

Obtained funding: Herrmann.

Administrative, technical, or material support:

Herrmann, Kirov, Dacian, Brömsen, Sadoni.

Supervision: Jeppsson, Kirov, Charitos, Massberg, Doenst, Juchem, Hagl.

Other: Dacian.

Conflict of Interest Disclosures: Dr Jeppsson reported serving on advisory boards of and delivering presentations for AstraZeneca, Bayer, Werfen, LFB Biotechnologies, Pharmacosmos, Boehringer Ingelheim, and Novo Nordisk unrelated to this work. Dr Sadoni reported serving as a proctor for Boston Scientific unrelated to this work. Dr Hagl reported serving as an advisor to and delivering presentations for Abbott, Edwards Lifesciences, and Abiomed unrelated to this work. No other conflicts were reported.

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